STT 843: Multivariate Analysis

7. Principal Component Analysis (Chapter 8.2-8.3)

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Outline

1. Principal Components for Covariance Matrices with Special Structures
2. Summarizing Sample Variation by Principle Components
3. The number of Principal Components
4. Interpretation the Sample Principal Components

Result 8.1

Let $\boldsymbol{\Sigma}$ be the covariance matrix associated with the random vector $\boldsymbol{X}^{\mathsf{T}} = [X_1, X_2, \ldots, X_p]$. Let Σ have the eigenvalue-eigenvector pair $(\lambda_1, \mathbf{e}_1), (\lambda_2, \mathbf{e}_2), \ldots, (\lambda_p, \mathbf{e}_p)$ where $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p \geq 0$. Then the $\boldsymbol{ith}$ principal component is given by

$$Y_i = \mathbf{e}_i^{\mathsf{T}} \boldsymbol{X} = e_{i1} X_1 + e_{i2} X_2 + \cdots + e_{ip} X_p, i = 1, 2, \ldots, p$$

With these choices,

$$\mathrm{Var}\left(Y_i\right) = \mathbf{e}_i^{\mathsf{T}} \boldsymbol{\Sigma} \mathbf{e}_i = \lambda_i, i = 1, 2, \ldots, p$$
$$\mathrm{Cov}\left(Y_i, Y_k\right) = \mathbf{e}_i^{\mathsf{T}} \boldsymbol{\Sigma} \mathbf{e}_k = 0, i \neq k$$

If some λ_i are equal, the choices of corresponding coefficients vectors, $\mathbf{e}_i$, and hence Y_i are not unique.

Results 8.2

Let $\boldsymbol{X}^{\mathsf{T}} = [X_1, X_2, \ldots, X_p]$ have covariance matrix $\boldsymbol{\Sigma}$, with eigenvalue-eigenvector pairs $(\lambda_1, \mathbf{e}_1), (\lambda_2, \mathbf{e}_2), \ldots, (\lambda_p, \mathbf{e}_p)$ where $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p \geq 0$.
Let $Y_1 = \mathbf{e}_1^{\mathsf{T}}\boldsymbol{X}, Y_2 = \mathbf{e}_2^{\mathsf{T}}\boldsymbol{X}, \ldots, Y_p = \mathbf{e}_p^{\mathsf{T}}\boldsymbol{X}$ be the principal components. Then

$$\sigma_{11}+\sigma_{22}+\cdots+\sigma_{pp} = \sum_{i=1}^{p} \mathrm{Var}\left(X_i\right) = \lambda_1+\lambda_2+\cdots+\lambda_p = \sum_{i=1}^{p} \mathrm{Var}\left(Y_i\right)$$

Results 8.3

If $Y_1 = \mathbf{e}_1^\mathsf{T}\boldsymbol{X}, Y_2 = \mathbf{e}_2^\mathsf{T}\boldsymbol{X}, \ldots, Y_p = \mathbf{e}_p^\mathsf{T}\boldsymbol{X}$ are the principal components obtained from the covariance matrix $\boldsymbol{\Sigma}$, then

$$\rho_{Y_i,X_k} = \frac{e_{ik}\sqrt{\lambda_i}}{\sqrt{\sigma_{kk}}}, \quad i,k = 1,2,\ldots,p$$

are the correlation coefficients between the components Y_i and the variables X_k. Here $(\lambda_1, \mathbf{e}_1), (\lambda_2, \mathbf{e}_2), \ldots, (\lambda_p, \mathbf{e}_p)$ are the eigenvalue-eigenvector pair for $\boldsymbol{\Sigma}$.

Example 8.1

Suppose the random variables X_1, X_2 and X_3 have the covariance matrix

$$\mathbf{\Sigma} = \begin{bmatrix} 1 & -2 & 0 \\ -2 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Calculating the population principal components

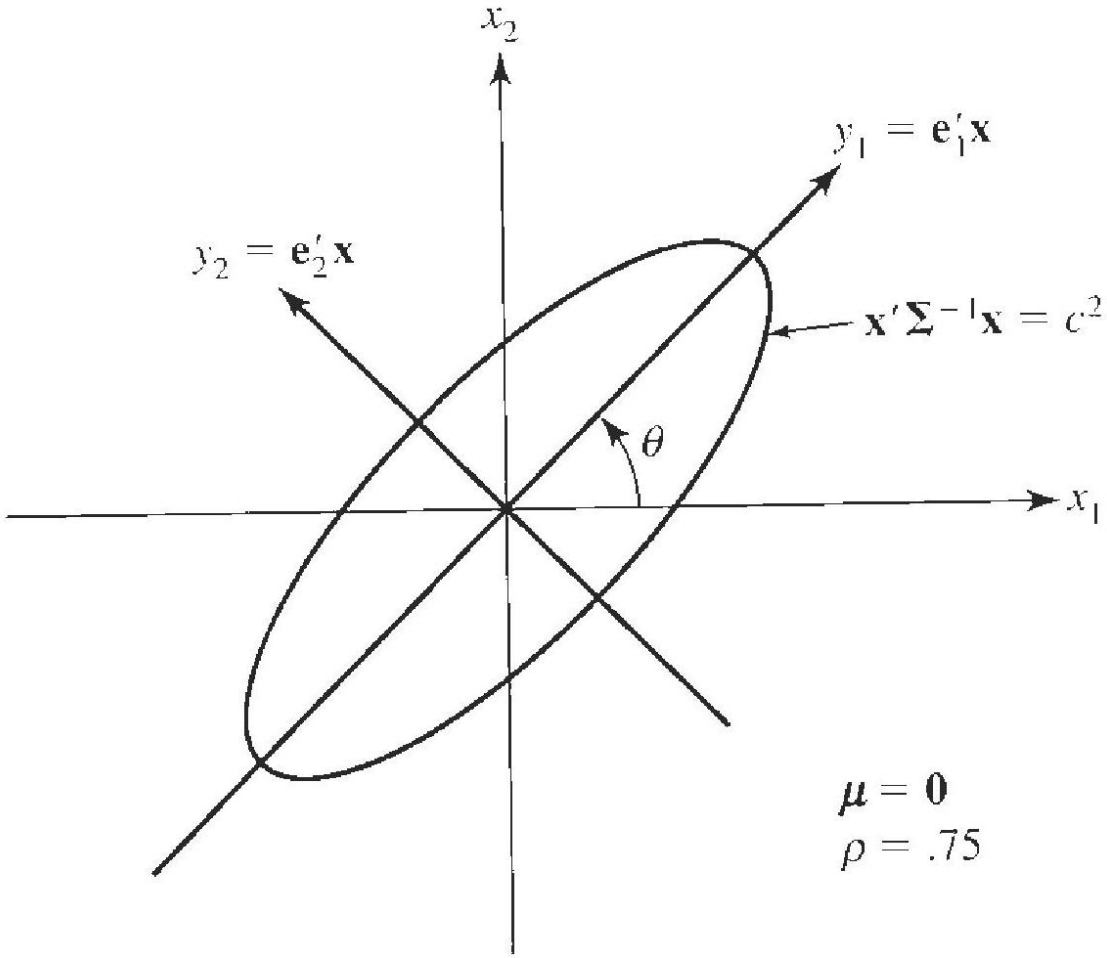


Figure 8.1 The constant density ellipse $\mathbf{x}^\mathsf{T}\Sigma^{-1}\mathbf{x} = c^2$ and the principal components y_1, y_2 for a bivariate normal random vector $\mathbf{X}$ having mean 0.

Suppose $\boldsymbol{X}$ is distributed as $N_p(\boldsymbol{\mu}, \Sigma)$. We know that the density of $\boldsymbol{X}$ is constant on the $\boldsymbol{\mu}$ centered ellipsoids

$$(\mathbf{x} - \boldsymbol{\mu})^\mathsf{T} \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) = c^2$$

which have axes $\pm\sqrt{\lambda_i}\mathbf{e}_i, i = 1, 2, \ldots, p$, where the $(\lambda_i, \mathbf{e}_i)$ are the eigenvalue eigenvector pairs of Σ.

Assume $\boldsymbol{\mu} = 0$, the equation above can be rewritten as

$$c^2 = \mathbf{x}^\top \boldsymbol{\Sigma}^{-1} \mathbf{x} = \frac{1}{\lambda_1} \left(\mathbf{e_1}^\top \mathbf{x}\right)^2 + \frac{1}{\lambda_2} \left(\mathbf{e}_2^\top \mathbf{x}\right)^2 + \cdots + \frac{1}{\lambda_p} \left(\mathbf{e}_p^\top \mathbf{x}\right)^2$$
$$= \frac{1}{\lambda_1} y_1^2 + \frac{1}{\lambda_2} y_2^2 + \cdots + \frac{1}{\lambda_p} y_p^2$$

where $\mathbf{e}_1^\top \mathbf{x}, \mathbf{e}_2^\top \mathbf{x}, \ldots, \mathbf{e}_p^\top \mathbf{x}$ are recognized as the principal components of $\mathbf{x}$. The equation above defines in a coordinate system with axes $y_1, y_2, \ldots, y_p$ lying in the direction $\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_p$, respectively.

Principal Components Obtained from Standardized Variables

Principal components may also be obtained from the standardized variables

$$Z_i = \frac{X_i - \mu_i}{\sqrt{\sigma_{ii}}}, i = 1, 2, \ldots, p$$

Or in matrix notation $\mathbf{Z} = \left(\mathbf{V}^{1/2}\right)^{-1}(\boldsymbol{X} - \boldsymbol{\mu})$. Clearly $\mathrm{E}(\mathbf{Z}) = 0$ and $\mathrm{Cov}(\mathbf{Z}) = \left(\mathbf{V}^{1/2}\right)^{-1} \Sigma \left(\mathbf{V}^{1/2}\right)^{-1} = \boldsymbol{\rho}$

Results 8.4

The i th principal component of the standardized variables $\mathbf{Z}^{\mathsf{T}} = [Z_1, Z_2, \ldots, Z_p]$ with $\mathrm{Cov}(\mathbf{Z}) = \boldsymbol{\rho}$, is given by

$$Y_i = \mathbf{e}_i^{\mathsf{T}} \mathbf{Z} = \mathbf{e}_i \left(\mathbf{V}^{1/2}\right)^{-1} (\boldsymbol{X} - \boldsymbol{\mu}), i = 1, 2, \ldots, p$$

Moveover,

$$\sum_{i=1}^{p} \mathrm{Var}\left(Y_i\right) = \sum_{i=1}^{p} \mathrm{Var}\left(Z_i\right) = p$$

and

$$\rho_{Y_i, Z_k} = e_{ik}\sqrt{\lambda_i}, i, k = 1, 2, \ldots, p$$

In this case $\left(\lambda_1, \mathbf{e}_1\right), \left(\lambda_2, \mathbf{e}_2\right), \ldots, \left(\lambda_p, \mathbf{e}_p\right)$ are the eigenvalue-eigenvector pairs for ρ, with $\lambda_1 \geq \lambda_2 \geq \cdots \lambda_p \geq 0$.

Example 8.2

Consider the covariance matrix

$$\boldsymbol{\Sigma} = \begin{bmatrix} 1 & 4 \\ 4 & 100 \end{bmatrix} \quad \text{and} \quad \boldsymbol{\rho} = \begin{bmatrix} 1 & .4 \\ .4 & 1 \end{bmatrix}$$

Obtain the principle components by the covariance matrix Σ and correlation matrix ρ.

Principal Components for Covariance Matrices with Special Structures

1. $$\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_{11} & 0 & \cdots & 0 \\ 0 & \sigma_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_{pp} \end{bmatrix}$$

2. $$\boldsymbol{\Sigma} = \begin{bmatrix} \sigma^2 & \rho\sigma^2 & \cdots & \rho\sigma^2 \\ \rho\sigma^2 & \sigma^2 & \cdots & \rho\sigma^2 \\ \vdots & \vdots & \ddots & \vdots \\ \rho\sigma^2 & \rho\sigma^2 & \cdots & \sigma^2 \end{bmatrix} \quad \text{or} \quad \boldsymbol{\rho} = \boldsymbol{\Sigma} = \begin{bmatrix} 1 & \rho & \cdots & \rho \\ \rho & 1 & \cdots & \rho \\ \vdots & \vdots & \ddots & \vdots \\ \rho & \rho & \cdots & 1 \end{bmatrix}$$

Summarizing Sample Variation by Principle Components

Suppose the data $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$ represent n independent drawings from some p-dimensional population withe mean vector $\boldsymbol{\mu}$ and covariance matrix Σ. These data yield the sample mean vector $\bar{\mathbf{x}}$, the sample covariance matrix $\mathbf{S}$, and the sample correlation matrix $\mathbf{R}$.

If $\mathbf{S} = s_{ik}$ be $p \times p$ sample covariance matrix with eigenvalue-eigenvector pairs $\left(\hat{\lambda}_1, \hat{\mathbf{e}}_1\right), \left(\hat{\lambda}_2, \hat{\mathbf{e}}_2\right), \ldots, \left(\hat{\lambda}_p, \hat{\mathbf{e}}_p\right)$, the ith sample principal component is given by

$$\hat{y}_i = \hat{\mathbf{e}}_i^{\mathsf{T}} \mathbf{x} = \hat{e}_{i1} x_1 + \hat{e}_{i2} x_2 + \cdots + \hat{e}_{ip} x_p, i = 1, 2, \ldots, p$$

where $\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \cdots \geq \hat{\lambda}_p \geq 0$ and $\mathbf{x}$ is any observation on the variables $\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_p$.

Also

$$\text{Sample variance } (\hat{y}_k) = \hat{\lambda}_k, k = 1, 2, \ldots, p$$

Sample covariance $(\hat{y}_i, \hat{y}_k) = 0, i \neq k$

$$\text{Total sample variance } = \sum_{i=1}^{n} s_{ii} = \hat{\lambda}_1 + \hat{\lambda}_2 + \cdots + \hat{\lambda}_p$$

$$r_{\hat{y}_i, x_k} = \frac{\hat{e}_{ik}\sqrt{\hat{\lambda}_i}}{\sqrt{s_{kk}}}, i, k = 1, 2, \ldots, p.$$

Example (Summarizing sample variability with two sample principal components)

A census provided information, by tract, on five socioeconomic variables for Madison, Wisconsin, area. The data from 61 tracts are list in Table 8.5. These data produced the following summary statistics

$\bar{\mathbf{x}}^{\mathsf{T}} = [4.47,$	$3.96,$	$71.42,$	$26.91,$	$1.64]$
total	professional	employed population	degree age over 16 employment	median home value
(thousands)	(percent)	(percent)	(percent)	$\$100,000$

and

$$\mathbf{S} = \begin{bmatrix} 3.397 & -1.102 & 4.306 & -2.078 & 0.027 \\ -1.102 & 9.673 & -1.513 & 10.953 & 1.203 \\ 4.306 & -1.513 & 55.626 & -28.937 & -0.044 \\ -2.078 & 10.953 & -28.937 & 89.067 & 0.957 \\ 0.027 & 1.203 & -0.044 & 0.957 & 0.319 \end{bmatrix}$$

Can the sample variation be summarized by one or two principal components?

Coefficients for the Principal Components (Correlation Coefficients in Parentheses)

Variable	$\hat{\mathbf{e}}_1\ (r_{\hat{y}_1,x_k})$	$\hat{\mathbf{e}}_2\ (r_{\hat{y}_2,x_k})$	$\hat{\mathbf{e}}_3$	$\hat{\mathbf{e}}_4$	$\hat{\mathbf{e}}_5$
Total population	$-0.039(-.22)$	0.071(.24)	0.188	0.977	-0.058
Profession	0.105(.35)	0.130(.26)	-0.961	0.171	-0.139
Employment (%)					
Government employment (%)	$-0.492(-.68)$	0.864(.73)	0.046	-0.091	0.005
Medium home value	0.863(.95)	0.480(.32)	0.153	-0.030	0.007
Variance $\left(\hat{\lambda}_i\right)$:	107.02	39.67	8.37	2.87	0.15
Cumulative percentage of total variance	67.7	92.8	98.1	99.9	1.000

The number of Principal Components

There is always the question of how many components to retain. There is no definitive answer to this question. Things to consider include

- the amount of total sample variance explained,
- the relative sizes of the eigenvalues (the variances of the sample components,)
- the subject-matter interpretation of the components.
- In addition, a component associated with an eigenvalue near zero and, hence deemed unimportant, may indicate an unsuspected linear dependency in the data.

A useful visual aid to determining an appropriate number of principal components is a scree plot. With the eigenvalues ordered from largest to smallest, a scree plot is a plot of $\hat{\lambda}_i$ versus i - the magnitude of an eigenvalue versus its number.

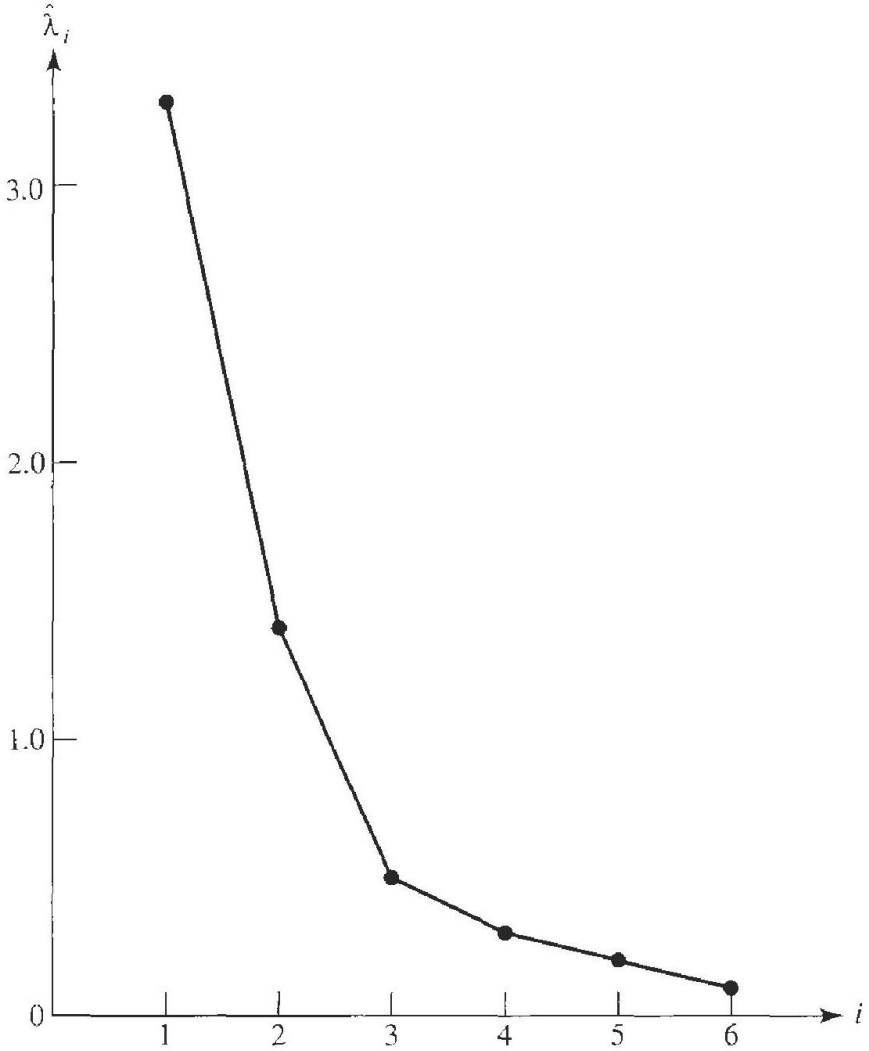


Figure 8.2 A scree plot.

Example 5.4 (Summarizing sample variability with one sample principal component)

In a study of size and shape relationships for painted turtles, Jolicoeur and Mosimann measured carapace length, width, and height. Their data, reproduced in Table 6.9 suggest an analysis in term s of logarithms (Jolicoeur generally suggests a logarithmic transformation in studies of size-andshape relationships). Perform a principal component analysis.

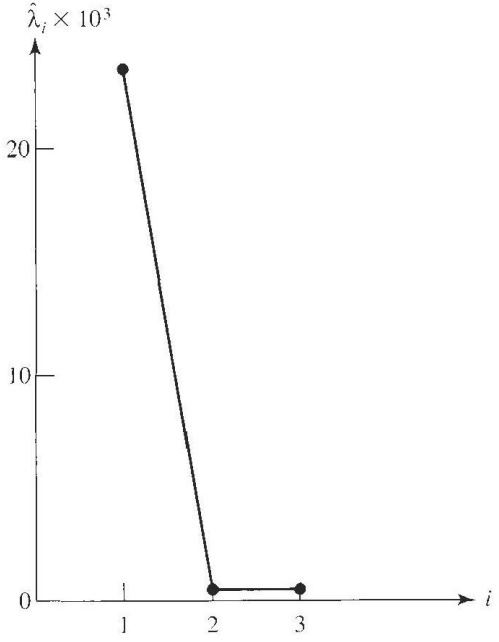


Figure 8.3 A scree plot for the turtle data.

Coefficients for the Principal Components (Correlation Coefficients in Parentheses)

Variable	$\hat{\mathbf{e}}_1$ $(r_{\hat{y}_1,x_k})$	$\hat{\mathbf{e}}_2$	$\hat{\mathbf{e}}_3$
ln (length)	.683(.99)	-.159	-.713
ln (width)	.510(.97)	-.594	.622
ln (height)	.523(.97)	.788	.324
Variance $\left(\hat{\lambda}_i\right)$:	23.30×10^{-3}	$.60 \times 10^{-3}$	$.36 \times 10^{-3}$
Cumulative percentage of total variance	96.1	98.5	100

Interpretation the Sample Principal Components

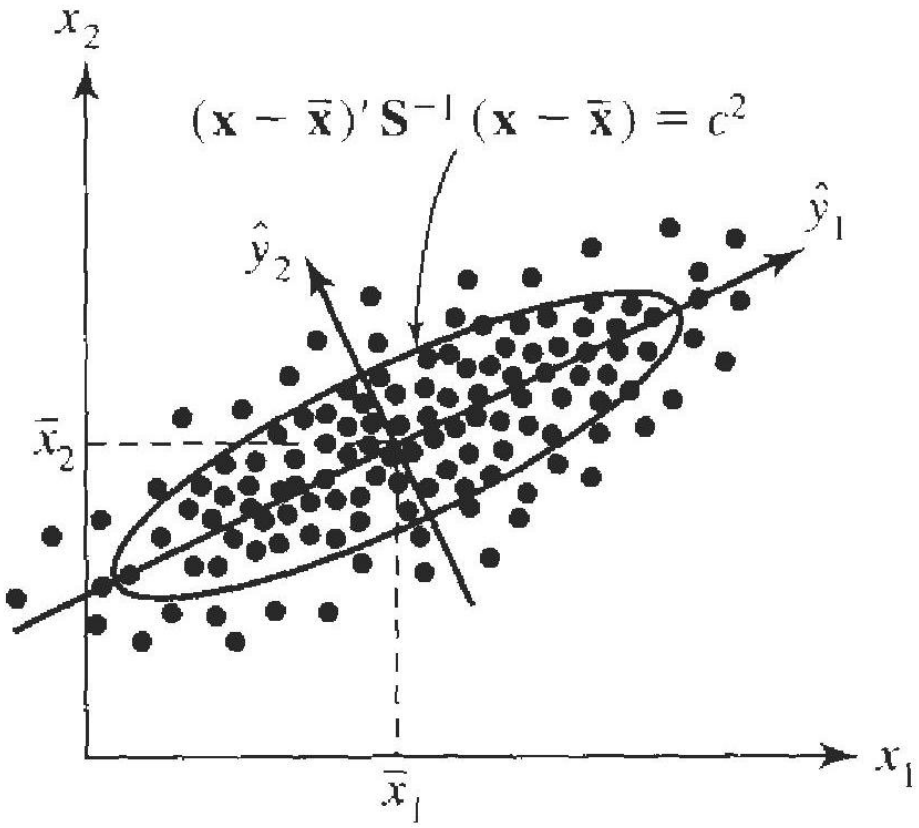
The sample principal components have several interpretations

- Suppose the underlying distribution of $\boldsymbol{X}$ is nearly $N_p(\boldsymbol{\mu}, \Sigma)$, Then the sample principal components $\hat{y}_i = \hat{\mathbf{e}}_i^{\mathsf{T}}(\mathbf{x} - \overline{\mathbf{x}})$ are realizations of population principal components $Y_i = \mathbf{e}_i(\boldsymbol{X} - \boldsymbol{\mu})$, which have an $N_p(0, \Lambda)$ distribution. The diagonal matrix Λ has entries $\lambda_1, \lambda_2, \ldots, \lambda_p$ and $(\lambda_i, \mathbf{e}_i)$ are the eigenvalue-eigenvector pairs of $\boldsymbol{\Sigma}$.

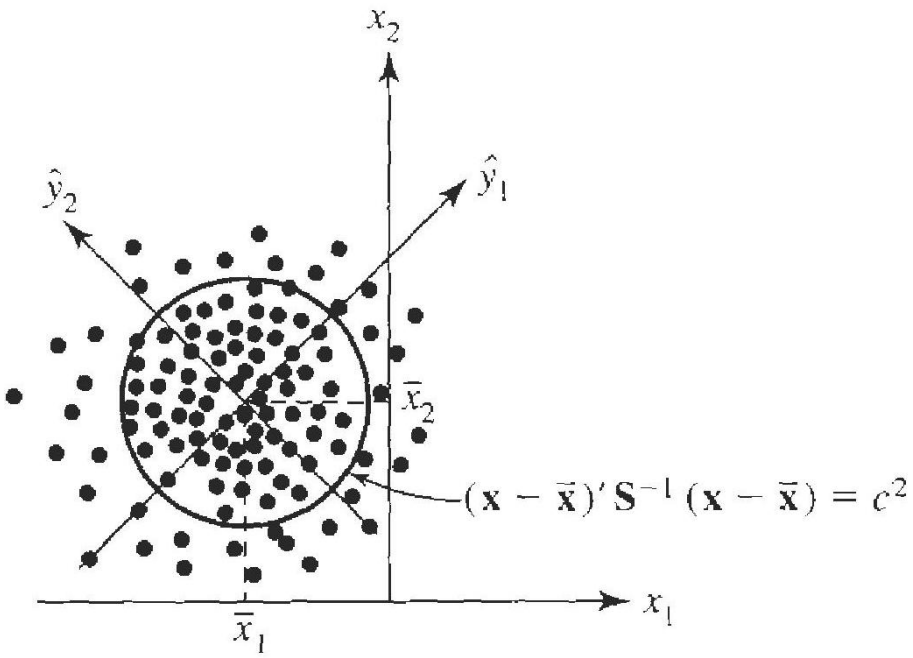
- From the sample value $\mathbf{x}_j$, we can approximate $\boldsymbol{\mu}$ by $\bar{\mathbf{x}}$ and Σ by $\mathbf{S}$. If $\mathbf{S}$ is positive definite, the contour consisting of all $p \times 1$ vector $\mathbf{x}$ satisfying

$$(\mathbf{x} - \bar{\mathbf{x}})^\top \mathbf{S}^{-1} (\mathbf{x} - \bar{\mathbf{x}}) = c^2$$

  estimates the constant density contour $(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) = c^2$ of the underlying normal density.
- Even when the normal assumption is suspect and the scatter plot may depart somewhat from an elliptical pattern, we can still extract eigenvalues from $\mathbf{S}$ and obtain the sample principal components.



(a) $\hat{\lambda}_1 > \hat{\lambda}_2$



(b) $\hat{\lambda}_1 \doteq \hat{\lambda}_2$

Sample principal components and ellipses of constant distance.

- Sample principal components are, in general, not invariant with respect to changes in scale.
- Variables measured on different scales or on a common scale with widely differing ranges are often standardized. For example, standardization is accomplished by constructing

$$\mathbf{z}_j = \mathbf{D}^{-1/2}(\mathbf{x}_j - \bar{\mathbf{x}}) = \left[\frac{x_{j1} - \bar{x}_1}{\sqrt{s_{11}}}, \frac{x_{j2} - \bar{x}_2}{\sqrt{s_{22}}}, \ldots, \frac{x_{j1} - \bar{x}_p}{\sqrt{s_{pp}}}\right]^{\mathsf{T}},$$

j=1,2, . . . , n

If $\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_n$ are standardized observations with covariance matrix $\mathbf{R}$, the i th sample principal component is

$$\hat{y}_i = \hat{\mathbf{e}}_i^{\mathsf{T}} \mathbf{z} = \hat{e}_{i1} z_1 + \hat{e}_{i2} z_2 + \cdots + \hat{e}_{ip} z_p, i = 1, 2, \ldots, p$$

where $\left(\hat{\lambda}_i, \hat{\mathbf{e}}_i\right)$ is the i th eigenvalue-eigenvector pair of $\mathbf{R}$ with

$$\text{Sample variance } (\hat{y}_i) = \hat{\lambda}_i, i = 1, 2, \ldots, p$$

Sample covariance $(\hat{y}_i, \hat{y}_k) = 0, i \neq k$

In addition,

Total (standardized) sample variance $= \text{tr}(\mathbf{R}) = p = \hat{\lambda}_1 + \hat{\lambda}_2 + \cdots + \hat{\lambda}_p$

and

$$r_{\hat{y}_i, z_k} = \hat{e}_{ik}\sqrt{\hat{\lambda}_i}, i, k = 1, 2, \ldots, p$$